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ITA-408 Ethical Hacking CO1- Assessment tool-3

- 1) You Discover a Vulnerability So Severe that Exploiting it even briefly could Crash Critical hospital Patient monitoring System You Contract Requires a quote Full Exploitation to Confirm Impact a quote. How do You Ethically resolve this Conflict between Contractual obligation and Patient Safety??

A) Ethical Resolution of a Critical Hospital Vulnerability During Penetration Testing

Introduction:

Ethical Hacking Plays a vital Role in Protecting Organizations From Cyberattacks. By identifying and Fixing Security Weakness before malicious hackers can exploit them Ethical hackers are hired to Performs Penetration Testing under Legal Argument with Usually define the Scope of testing than can be Used However these are Situations Followings the Contract may create Serious ethical

Concerns:-

Consider the Following Scenario:-

"You Discover a vulnerability a Server that Exploiting it even briefly Crash and Critical hospital Patient monitoring Systems You Contract Full exploitation to Confirm Impact. How do You Ethically resolve this Conflict Between Contracted obligation and Patient Safety..."

This Situation Create a Conflict Between Contractual responsibility and the Ethical duty to protect human life. The Best Ethical decision is prioritize patient safety while communicating professionally with the client and documenting the finding responsibly.

Understanding the Ethical Conflict...

The Penetration testing Contract Requires the tester to perform Full Exploration to demonstrate Actual impact of Vulnerability. Normally this helps the client understand the seriousness of Security Issue...

However this case, exploiting the Vulnerability could:

- Crash Patient-monitoring Systems.
- Intercept Life Saving medical Services
- Delay Emergency treatment
- Put Patients lives at immediate Risk
- Causes Legal and Financial Consequences For the Hospital.

Therefore blindly Following the Contract violate the most important ethical Principle Do no harm...

An Ethical Hacker's Responsibility extends beyond technical testing. They must Ensure that their actions do not Create Unnecessary risks For people Especially in healthcare Environments.

Ethical Principles Involved...

- 1) Several Professional Ethical Principles Apply in the Situation.
If exploration could interrupt critical medical equipment the Ethical Hacker must avoid actions that could endanger patients.
Savings lives always take priority over demonstrating technical proof.

- 2) Minimise Harm:

One of the primary of Ethical Hacking is to improve security not to damage systems.

Ethical Hacker should never creates risk who safer alternative exists.

Instead of Exploiting of vulnerability on a live hospital system they should collect enough evidence to demonstrate the issue safely...

- 3) Act Responsibility:

responsible cyber security professional the technical capability does not justify - Unsafe Actions.

Even if permission exist the contract professionals must exercise judgement and avoid actions that could server consequence.

Advantages of the Ethical Approach:-

Following this Ethical Process provide many benefits.

Protects Patients:-

No interruption to medical treatment

Maintains Trust:-

The hospital continues to trust the Security team.

Supports Responsible Disclosure:-

The vulnerability is reported privately and fixed quickly.

Reduces Legal Risk:-

Avoids negligence and liability.

Maintain Professional Reputation

Demonstrate responsible Cyber Security Practice.

Improve Security:-

The vulnerability is fixed without creating additional damage.

Conclusion:-

Instead of ethical Hacker immediately stop testing documents to vulnerability notify the hospital recommended the safe testing method and assist with remediation professional ethical legal responsibility and human safety support their decision.

Ultimately responsibility of Ethical Hacker is not simply contact has to protect prevent harm act with integrity..